

Capstone Project Guidelines – BRICS Astronomy/IDIA Data Analytics

Objective

The capstone project is an opportunity for participants to apply their data analytics skills to a research or real-world problem, ideally using astronomy-related data but open to other domains. Projects should demonstrate proficiency in data exploration, visualization, modeling (if applicable), and interpretation of results.

Project Requirements

1. Project Topic

- Should involve a dataset that allows for meaningful analysis or insights.
- Can be astronomy-focused or from another domain such as climate, health, or economics.
- Participants are encouraged to select a topic of personal or regional relevance within the BRICS context.

2. Submission Format

- Submit a single **Google Colab notebook** containing all code, explanations, visualizations, and conclusions.
- Ensure the notebook is well-commented and can be followed by someone with no prior context.
- Share your final notebook as a **view-only link** (no editing permissions).

3. Notebook Structure

- Title & Objective
- Dataset Description
- Data Cleaning/Preprocessing
- Exploratory Data Analysis (EDA)
- Modeling/Analysis (if applicable)
- Results & Interpretation
- Conclusion & Future Work
- References/Data Sources

Timeline

- You will have **2–3 weeks** to complete the project.
- Final deadline: Week 8



Suggested Data Sources

Astronomy & Space Science

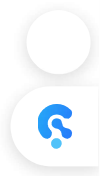
- [ESA Gaia Archive](#)
- [NASA Exoplanet Archive](#)
- [Astroquery \(Python\)](#)
- [Vizier](#)

Other Open Datasets

- [Kaggle Datasets](#)
- [World Bank Open Data](#)
- [Our World in Data](#)
- [Google Dataset Search](#)
- [UCI Machine Learning Repository](#)

Grading Rubric (100 points total)

Component	Points	Description
Problem Definition	10	Clearly defined research question and objectives.
Dataset Selection & Justification	10	Relevance and appropriateness of chosen dataset.
Data Cleaning & Preprocessing	15	Evidence of thoughtful preparation of data.
Exploratory Data Analysis (EDA)	15	Quality and depth of visualizations and insights.



Component	Points	Description
Modeling / Analysis	20	Appropriate use of analytical or ML methods (if applicable).
Interpretation & Results	15	Clarity and significance of insights drawn.
Code Quality & Reproducibility	10	Well-commented, organized, and reproducible notebook.
Presentation & Structure	5	Overall readability and flow of the notebook.

Bonus: Up to 5 **bonus points** may be awarded for creativity, novelty, or interdisciplinary relevance.

Submission

- Submit your **view-only Colab notebook link** via the course platform.
- Ensure the notebook **runs from top to bottom** without errors.

If you have questions or need help selecting a dataset, feel free to ask during the live sessions or in the course Slack forum.

